

Module 1 : Introduction to the cloud

Defining cloud computing

To review the definition of cloud computing, choose each of the following four numbered markers.



Cloud deployment types

You can deploy cloud resources in multiple ways: cloud, on-premises, and hybrid. Each type of deployment has its own benefits and considerations, and exploring these options can help you make informed decisions about your cloud strategy.

To learn more about cloud deployment types, choose each of the following flashcards.



Cloud : In a cloud-based deployment model, you have the flexibility to migrate your existing resources to the cloud, design and build new applications within the cloud environment, or use a combination of both.

For instance, a company might migrate data resources to the cloud, then develop an application comprised of virtual servers, databases, and networking components entirely hosted in the cloud.

On-premises : Deploying resources on premises using virtualization and resource management tools does not provide many of the benefits of cloud computing. However, it is sometimes sought for its ability to provide dedicated resources and low latency.

In most cases this deployment model is the same as legacy IT infrastructure while using application management and virtualization technologies to try increasing resource utilization.

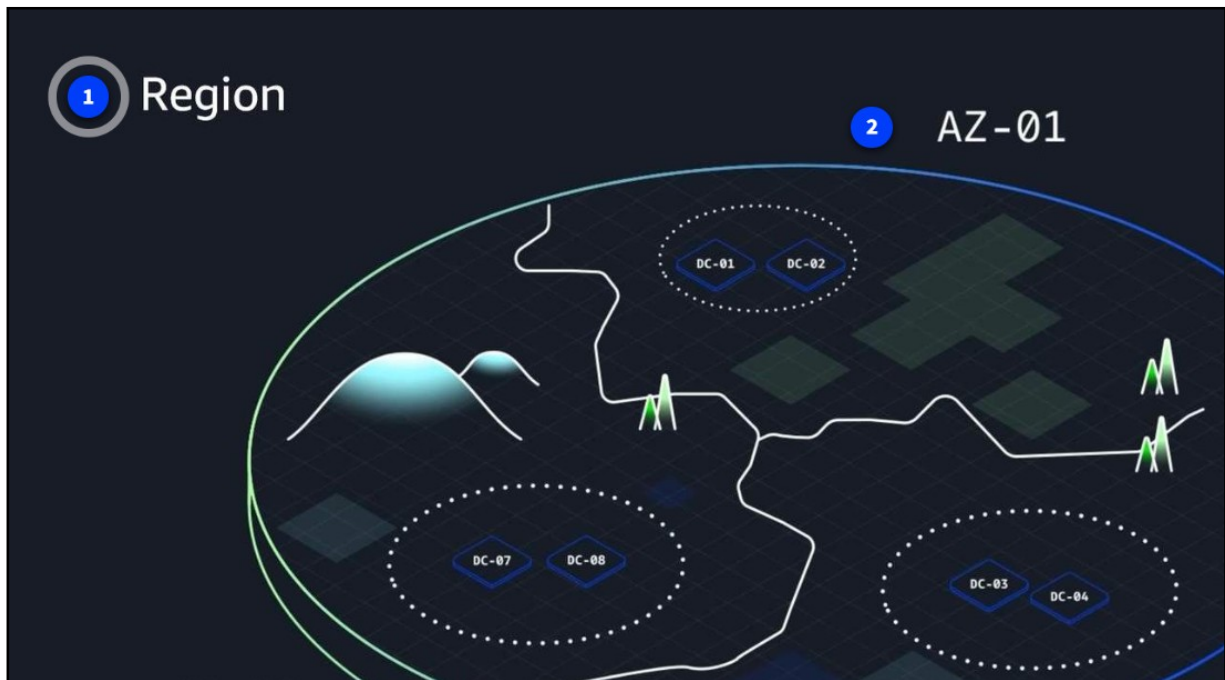
Hybrid : In a hybrid deployment, cloud-based resources and on-premises infrastructure work together. This approach is ideal for situations where legacy applications must remain on premises due to maintenance preferences or regulatory requirements.

For instance, a company might choose to retain certain regulated legacy applications on-premises while using cloud services for advanced data processing and analytics.

Multi-cloud deployments can also be considered *hybrid deployments*.

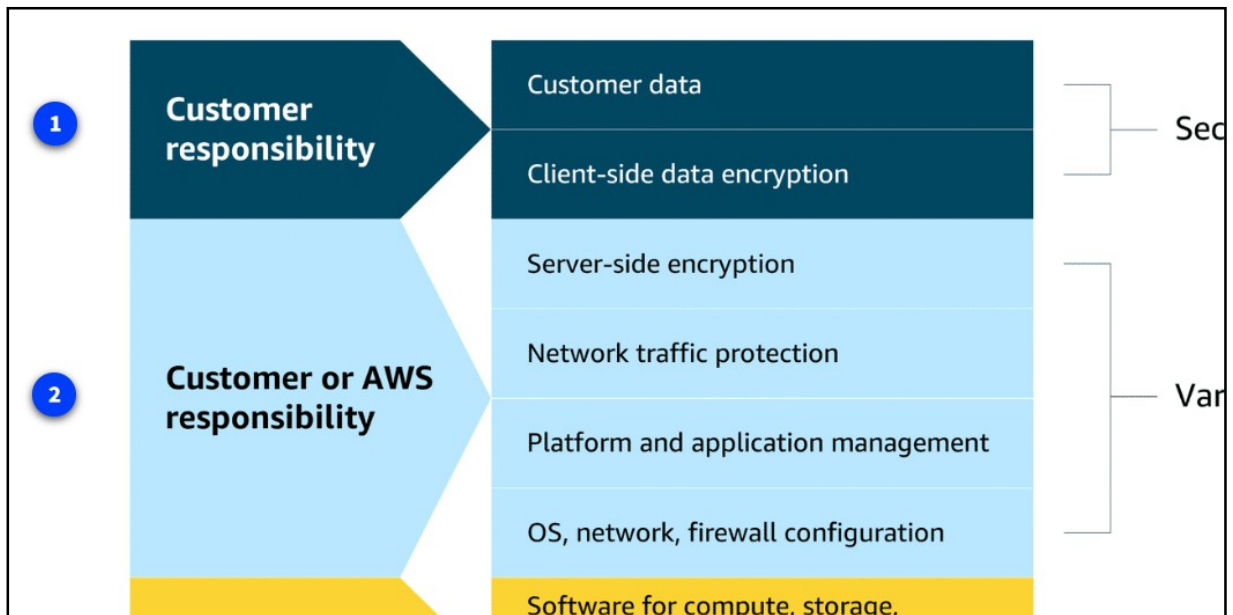
Key benefits of the AWS Cloud

- *Trade fixed expense for variable expense*
- *Benefit from massive economies of scale*
- *Stop guessing capacity*
- *Increase speed and agility*
- *Stop spending money to run and maintain data centers*
- *Go global in minutes*



AWS Regions : AWS Regions are physical locations around the world that contain groups of data centers. These groups of data centers are called Availability Zones. Each AWS Region consists of a minimum of three physically separate Availability Zones within a geographic area.

Availability Zones : An Availability Zone consists of one or more data centers with redundant power, networking, and connectivity. Regions and Availability Zones are designed to provide low-latency, fault-tolerant access to services for users within a given area.

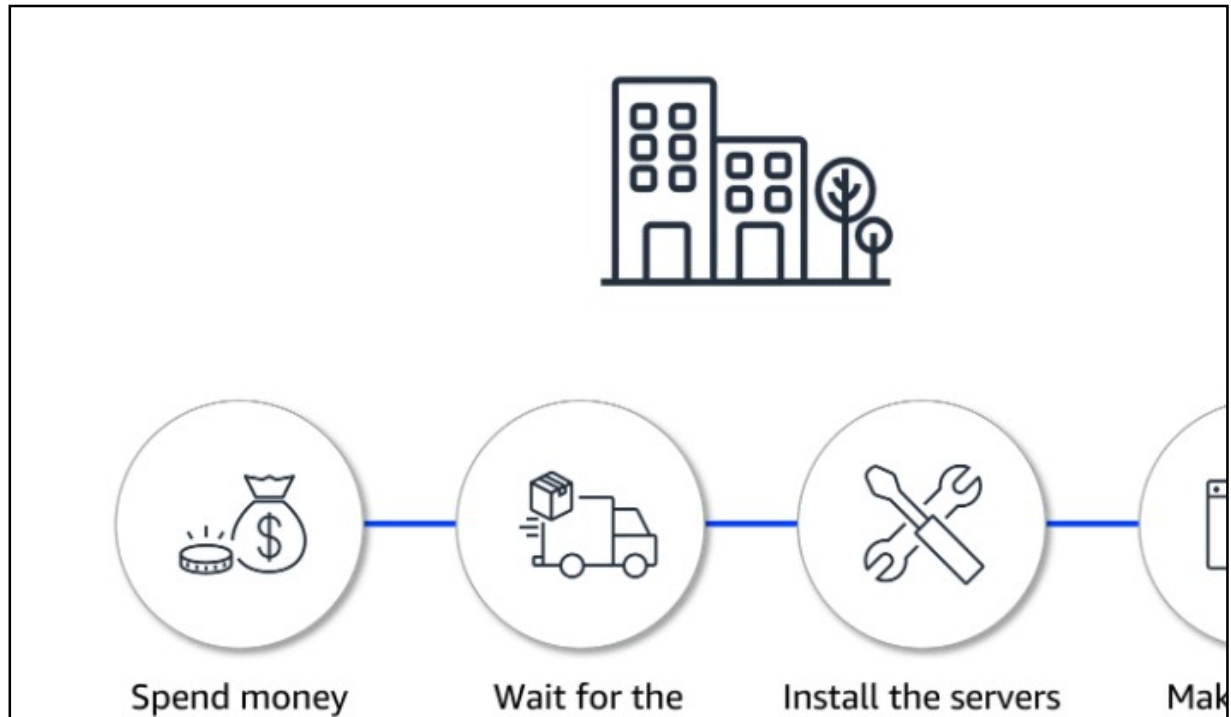


Real-Life cloud use cases:

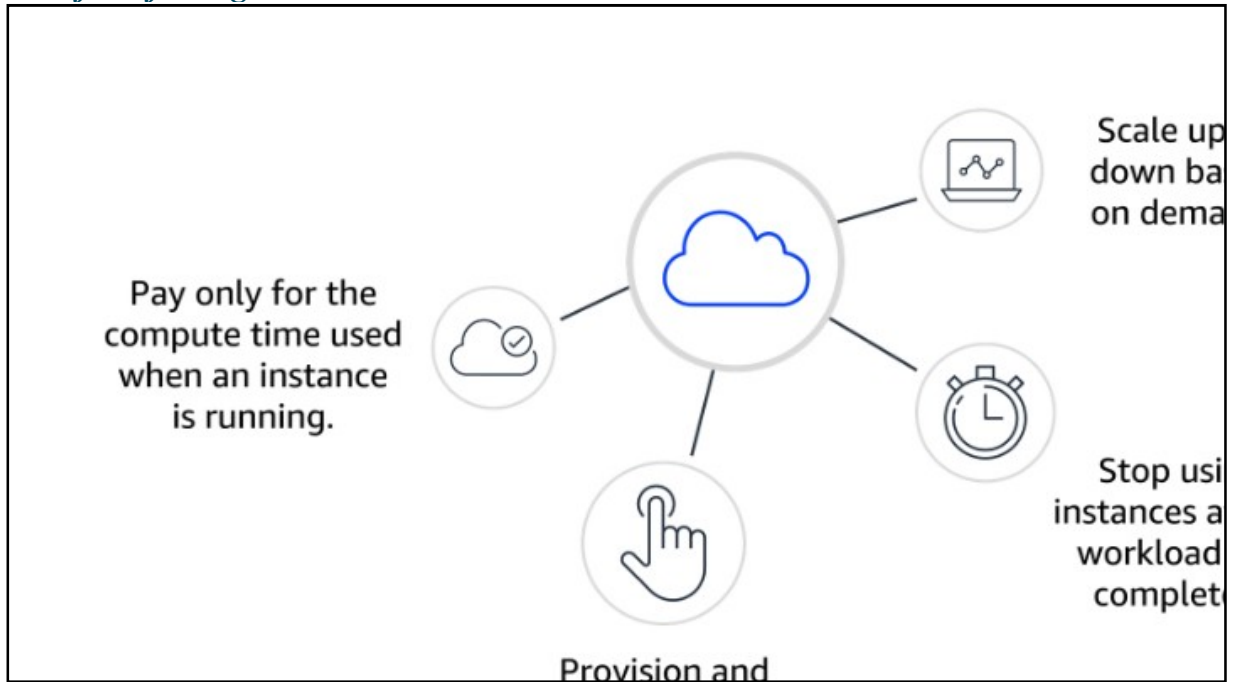
- This ecommerce company is based in Seattle, Washington in the United States. The company wants to expand to global locations. However, the further the computing infrastructure is from their customers, the longer the latency. The company decides to expand to global AWS Regions to better reach their global customers.
- The company deploys resources to the eu-west-1 Region in Ireland. Deploying in multiple Regions increases high availability. The company increases fault-tolerance and high availability even more by deploying to two Availability Zones in this Region. The company does not need to worry about securing the data center in Ireland because securing the physical infrastructure is an AWS responsibility. The company can instead focus on securing and encrypting their data within their cloud resources.
- The company has a significant customer base in Asia, too. So, they deploy resources to the ap-southwest-1 AWS Region in Singapore. Rather than having to set up physical infrastructure at an international scale, which can take months or years, the company used AWS to deploy global operations within a matter of minutes.

Module 2 – Compute in the Cloud

Challenges of on-premises resources



Benefits of using Cloud Resources



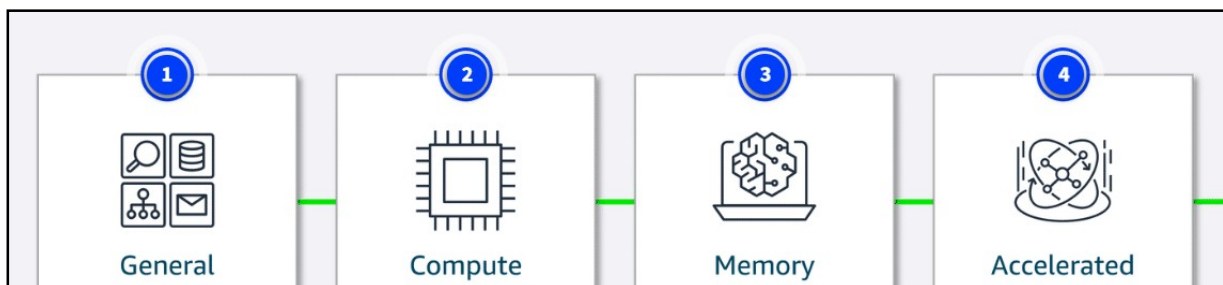
Amazon EC2

Launch -> Connect -> Use

Amazon EC2 is more flexible, cost-effective, and faster than managing on-premises servers. It offers on-demand compute capacity that can be quickly launched, scaled, and terminated, with costs based only on active usage.

The flexibility of Amazon EC2 allows for faster development and deployment of applications. You can launch as many or as few virtual servers as needed and configure security, networking, and storage. You can also scale resources up or down based on usage, such as handling high traffic or compute-heavy tasks.

EC2 instance types



- General purpose instances provide a balanced mix of compute, memory, and networking resources. They are ideal for diverse workloads, like web services, code repositories, and when workload performance is uncertain.
- Compute optimized instances are ideal for compute-intensive tasks, such as gaming servers, high performance computing (HPC), machine learning, and scientific modeling.
- Memory optimized instances are used for memory-intensive tasks like processing large datasets, data analytics, and databases. They provide fast performance for memory-heavy workloads.
- Accelerated computing instances use hardware accelerators, like graphics processing units (GPUs), to efficiently handle tasks, such as floating-point calculations, graphics processing, and machine learning.
- Storage optimized instances are designed for workloads that require high performance for locally stored data, such as large databases, data warehousing, and I/O-intensive applications.

How to provision AWS resources

The **AWS Management Console** is a web interface for managing AWS services, offering quick access to services, search functionality, and simplified workflows. With the mobile app, you monitor resources, view alarms, and check billing, supporting multiple logged-in identities at once.

Good for: Users who prefer a visual, easy-to-use interface for managing and configuring AWS services

With the **AWS CLI**, you manage multiple AWS services directly from the command line across Windows, macOS, and Linux. You can automate tasks through scripts, such as launching EC2 instances.

Good for: Advanced users and developers who need to automate tasks, script actions, and manage AWS resources efficiently from the command line

The **AWS SDK** simplifies integrating AWS services into your applications by providing APIs for various programming languages. AWS offers documentation and sample code for languages like C++, Java, and .NET to help you get started.

Good for: Developers looking to integrate AWS services into their applications using language-specific APIs

AWS pricing options

On-Demand Instances:

Pay only for the compute capacity you consume with no upfront payments or long-term commitments required.

Reserved Instances:

Get a savings of up to 75 percent by committing to a 1-year or 3-year term for predictable workloads using specific instance families and AWS Regions.

Spot Instances:

Bid on spare compute capacity at up to 90 percent off the On-Demand price, with the flexibility to be interrupted when AWS reclaims the instance.

Savings Plans:

Save up to 72 percent across a variety of instance types and services by committing to a consistent usage level for 1 or 3 years.

Dedicated Hosts:

Reserve an entire physical server for your exclusive use. This option offers full control and is ideal for workloads with strict security or licensing needs.

Dedicated Instances:

Pay for instances running on hardware dedicated solely to your account. This option provides isolation from other AWS customers.

Scaling Amazon EC2

Scalability

Scalability refers to the ability of a system to handle an increased load by adding resources. You can scale up by adding more power to existing machines, or you can scale out by adding more machines. Scalability focuses on long-term capacity planning to make sure that the system can grow and accommodate more users or workloads as needed.

Elasticity

Elasticity is the ability to automatically scale resources up or down in response to real-time demand. A system can then rapidly adjust its resources, scaling out during periods of high demand and scaling in when the demand decreases. Elasticity provides cost efficiency and optimal resource usage at any given moment.

Elastic Load Balancing

Elastic Load Balancing (ELB) automatically distributes incoming application traffic across multiple resources, such as EC2 instances, to optimize performance and reliability. A load balancer serves as the single point of contact for all incoming web traffic to an Auto Scaling group. As the number of EC2 instances fluctuates in response to traffic demands, incoming requests are first directed to the load balancer. From there, the traffic is distributed evenly across the available instances.

Although ELB and Amazon EC2 Auto Scaling are distinct services, they work in tandem to enhance application performance and ensure high availability. Together, they enable applications running on Amazon EC2 to scale effectively while maintaining consistent performance.

Benefits:

Efficient traffic distribution:

ELB evenly distributes traffic across EC2 instances, preventing overload on any single instance and optimizing resource utilization.

Automatic scaling:

ELB scales with traffic and automatically adjusts to changes in demand for a seamless operation as backend instances are added or removed.

Simplified management:

ELB decouples front-end and backend tiers and reduces manual synchronization. It also handles maintenance, updates, and failover to ease operational overhead.

Decoupling services

Monolithic applications

Applications consist of multiple components that work together to transmit data, fulfill requests, and keep the application running smoothly. In a traditional approach to application architecture, the components—such as database logic, web application servers, user interfaces, and business logic—are tightly coupled. This means that if one component fails, it can cause the failure of other components, potentially bringing down the entire application.

Microservices architecture

To improve application availability and resilience, you can adopt a microservices architecture. In this approach, application components are loosely coupled, meaning that if one component fails, the others continue to function normally. The communication between components remains intact, and the failure of a single component does not impact the entire system. This design promotes greater flexibility and reliability in the application.

Amazon EventBridge, Amazon SNS, and Amazon SQS are AWS services that help different parts of an application communicate effectively in the cloud.

EventBridge

EventBridge is a serverless service that helps connect different parts of an application using events, helping to build scalable, event-driven systems. With EventBridge, you route events from sources like custom apps, AWS services, and third-party software to other applications. EventBridge simplifies the process of receiving, filtering, transforming, and delivering events, so you can quickly build reliable applications.

Amazon SQS

Amazon SQS is a message queuing service that facilitates reliable communication between software components. It can send, store, and receive messages at any scale, making sure messages are not lost and that other services don't need to be available for processing. In Amazon SQS, an application places messages into a queue, and a user or service retrieves the message, processes it, and then removes it from the queue.

Amazon SNS

Amazon SNS is a publish-subscribe service that publishers use to send messages to subscribers through SNS topics. In Amazon SNS, subscribers can include web servers, email addresses, Lambda functions, and various other endpoints. You will learn about Lambda in more detail later.

Resource link	Description
Compute on AWS	This resource provides an overview of the different cloud computing services offered by AWS.
AWS Compute Blog	This blog provides updates, tutorials, and best practices for using AWS compute services, such as Amazon EC2, AWS Lambda, Amazon ECS, and more.
AWS Compute Services	This reference provides an in-depth introduction to the compute services available within the AWS Cloud.
Hands-On Tutorials: Compute	This resource provides practical, step-by-step tutorials designed to help users gain hands-on experience with AWS compute services. It is ideal for beginners and those new to cloud computing.
Amazon EC2	Amazon EC2 runs virtual servers in the cloud with flexible computing capacity.
Amazon EC2 Instance Types	This guide provides detailed information about the different types of EC2 instances, including their specifications, capabilities, and use cases. It helps you choose the right instance type based on your workload needs, such as compute, memory, and storage requirements.
Amazon EC2 Pricing	This guide explains the different pricing models for EC2 instances, including On-Demand, Reserved Instances, and Spot Instances, so you can choose the best option based on your usage.
Amazon EC2 Auto Scaling	Amazon EC2 Auto Scaling automatically adjusts instance count based on demand for high availability and cost-efficiency.
Elastic Load Balancing	Elastic Load Balancing automatically distributes incoming application traffic across multiple EC2 instances for high availability and fault tolerance.
Amazon Simple Notification Service	Amazon SNS is a messaging service for sending notifications to users or other applications through SMS, email, or mobile push notifications.
Amazon Simple Queue Service	Amazon SQS decouples application components through message queuing, storing and processing messages reliably.

Module 3 – Exploring Compute Services

Unmanaged and managed services

With unmanaged compute services like Amazon EC2, AWS takes care of the underlying physical infrastructure, but you're responsible for setting up, securing, and maintaining the operating system, network configurations, and applications on your instances. Managed services, on the other hand, reduce the amount of infrastructure you need to manage. While AWS handles much of the operational overhead, you might still need to perform some provisioning or configuration depending on the service.

Fully-managed services

Fully-managed services—like serverless ones—take abstraction even further, eliminating the need to provision or manage any servers at all. The underlying infrastructure is fully managed by AWS, so you can focus entirely on writing and deploying code. Later in this module, you will explore Lambda. Lambda is a serverless compute service where AWS handles the infrastructure, scaling, and availability, while you remain responsible for securing and managing your application code.

Lambda

Lambda is a serverless compute service that runs code in response to events without the need to provision or manage servers. It automatically manages the underlying infrastructure, scaling resources based on the volume of requests. You are charged only for the compute time consumed, down to the millisecond. Lambda handles execution, scaling, and resource allocation. You can optimize performance by configuring the appropriate memory size for your function.

Lambda use cases

Lambda is ideal for building responsive, event-driven applications across a wide range of industries. Here are three real-world examples that show how Lambda helps companies scale efficiently, reduce operational overhead, and only pay for what they use.

- **Real-time image processing for a social media application**
- **Personalized content delivery for a news aggregator**
- **Real-time event handling for an online game**

AWS container services

Amazon ECS

Amazon Elastic Container Service (Amazon ECS) is a scalable container orchestration service for running and managing containers on AWS, like Docker containers. Docker is a software platform for building, testing, and deploying applications quickly.

Amazon ECS launch types

- **Amazon ECS with Amazon EC2** is ideal for small-to-medium businesses that need full control over infrastructure. Suitable for custom applications requiring specific hardware or networking configurations, with the flexibility of Amazon EC2 and the simplicity of Amazon ECS.
- **Amazon ECS with AWS Fargate** is perfect for startups or small teams building web applications with variable traffic. It's a serverless option—no server management required—so teams can focus on development while Amazon ECS handles scaling and orchestration.

Amazon EKS

Amazon Elastic Kubernetes Service (Amazon EKS) is a fully managed service for running Kubernetes on AWS. It simplifies deploying, managing, and scaling containerized applications using open-source Kubernetes, with ongoing support and updates from the broader community.

Amazon EKS launch types

- **Amazon EKS with Amazon EC2:** This is best for enterprises needing full control over infrastructure. It offers deep customization of EC2 instances alongside Kubernetes scalability—ideal for complex, large-scale workloads.
- **Amazon EKS with AWS Fargate:** This is great for teams wanting Kubernetes flexibility without managing servers. It combines Kubernetes power with serverless simplicity, helping to scale applications quickly across various use cases.

Amazon ECR

Amazon Elastic Container Registry (Amazon ECR) is where you can store, manage, and deploy container images. It supports container images that follow the Open Container Initiative (OCI) standards. You can push, pull, and manage images in your Amazon ECR repositories using standard container tooling and command line interfaces (CLIs).

Fargate

AWS Fargate is a serverless compute engine for containers. It works with both Amazon ECS and Amazon EKS. Fargate is a container hosting platform, unlike Amazon ECS and Amazon EKS, which are both container orchestration services.

When using Fargate, you do not need to provision or manage servers. Fargate manages your server infrastructure for you. You can focus more on innovating and developing your applications, and you pay only for the resources that are required to run your containers.

Elastic Beanstalk

Elastic Beanstalk is a fully managed service that streamlines the deployment, management, and scaling of web applications. Developers can upload their code, and Elastic Beanstalk automatically handles the provisioning of infrastructure, scaling, load balancing, and application health monitoring. It supports various programming languages and frameworks, such as Java,

.NET, Python, Node.js, Docker, and more. It provides full control over the underlying AWS resources while automating many operational tasks.

Good for: Deploying and managing web applications, RESTful APIs, mobile backend services, and microservices architectures, with automated scaling and simplified infrastructure management

AWS Batch

AWS Batch is a fully managed service that you can use to run batch computing workloads on AWS. It automatically schedules, manages, and scales compute resources for batch jobs, optimizing resource allocation based on job requirements.

Good for: Processing large-scale, parallel workloads in areas like scientific computing, financial risk analysis, media transcoding, big data processing, machine learning training, and genomics research

Lightsail

Amazon Lightsail is a cloud service offering virtual private servers (VPSs), storage, databases, and networking at a predictable monthly price. It's ideal for small businesses, basic workloads, and developers seeking a straightforward AWS experience without the complexity of the full AWS Management Console.

Good for: Basic web applications, low-traffic websites, development and testing environments, small business websites, blogs, and learning cloud services

Outposts

AWS Outposts is a fully managed hybrid cloud solution that extends AWS infrastructure and services to on-premises data centers. It provides a consistent experience between on premises and the AWS Cloud, offering compute, storage, and networking components.

Good for: Low-latency applications, data processing in remote locations, migrating and modernizing legacy applications, and meeting regulatory compliance or data residency requirements

Resource link	Description
Containers on AWS	The AWS Containers Services page provides an overview of the AWS container offerings, including services for container image storage, orchestration, and compute. These offerings are designed to streamline the deployment and management of containerized applications.
Amazon Elastic Container Registry	The Amazon ECR is a fully managed service for storing, managing, and deploying container images securely at scale.
Amazon Elastic Container Service	Amazon ECS is a fully managed service that streamlines the deployment, management, and scaling of containerized applications on AWS.
Amazon Elastic Kubernetes Service	Amazon EKS is a fully managed Kubernetes service that streamlines running Kubernetes clusters on AWS and on premises. It automates infrastructure management and integrates with AWS networking, security, and storage services.
AWS Fargate	Fargate is a serverless compute engine for running containers without managing servers. It is integrated with Amazon ECS and Amazon EKS.
AWS Elastic Beanstalk	Elastic Beanstalk is a fully managed service for deploying and scaling web applications without managing infrastructure.
AWS Batch	AWS Batch is a fully managed service for efficiently running large-scale batch computing jobs on AWS.
What is Amazon Lightsail?	Lightsail is a simplified cloud platform offering VPS, containers, and databases with predictable pricing.
What is AWS Outposts?	AWS Outposts extends AWS infrastructure and services to on-premises locations for low-latency, local data processing.
Choosing a modern application strategy	The AWS Decision Guide for Modern Application Strategy helps organizations determine the most suitable development approach—serverless or Kubernetes—based on their operational model, team structure, and workload requirements.

Module 4 – Going Global

Key considerations when choosing Regions

Compliance

Compliance is an important consideration when selecting Regions for deploying business resources. Different geographical locations have varying regulatory requirements and data protection laws that organizations must follow. For example, the General Data Protection Regulation (GDPR) is designed to protect the personal data and privacy of individuals within the European Union (EU). An online retail company operating in the EU would be required to meet GDPR compliance to protect customer data. GDPR compliance includes obtaining proper consent for data collection and providing mechanisms for data access and deletion.

Proximity

When selecting a Region, you also want to consider how to achieve low latency for your users. Regions closer to your user base minimize data travel time, which reduces latency and enhances application responsiveness. Choosing a Region or set of Regions farther away from customers could introduce delays, which might impact user satisfaction and overall system efficiency.

Feature availability

You also want to consider which specific features and services are available in each Region. AWS is constantly expanding features and services to multiple locations, but not all Regions contain all AWS offerings. For example, AWS GovCloud Regions are specifically designed to meet the compliance and security requirements of US government agencies and their contractors. These Regions have stringent physical, operational, and personnel security controls in place. These controls are only available in specific Regions to meet certain governmental regulatory requirements.

Pricing

When selecting a Region, pricing is also a factor that can influence your decision. Some Regions have lower operational costs than others. These operational costs can impact the overall expenses for hosting applications and services. Tax laws and regulations can also play a role in cost. Some Regions might offer tax incentives or have lower tax rates, which can affect customer pricing. Additionally, data sovereignty laws in certain Regions might require data to be stored locally, affecting both compliance and cost.

Deploying multi-Region and multi-AZ resources

High availability: High availability refers to the capability of a system to operate continuously without failing. In the context of AWS infrastructure, it means that your applications can handle the failure of individual components without significant downtime.

Agility: Agility refers to the ability to quickly adapt to changing requirements or market conditions. With AWS infrastructure in place, you can modify and deploy services rapidly.

Elasticity: Elasticity refers to the ability of a system to scale resources up or down automatically in response to changes in demand. AWS infrastructure is set up for you to scale resources up and down on demand.

Edge locations

In addition to AWS Regions that contain Availability Zones, AWS has a global edge network that provides quicker content access to users outside of standard Regions. These edge locations are strategically placed in areas like Atlanta, Georgia, USA or Shanghai, China to provide low-latency access to AWS services and content delivery. Edge locations offer multiple services to run closer to end users, including AWS networking services like Amazon CloudFront. CloudFront is a content delivery network (CDN) and caching system that you learn more about later in this training.

AWS Regions

Regions are geographical areas around the world that are made up of multiple data centers. These data centers provide scalable and redundant infrastructure for hosting cloud services. Each Region consists of multiple, isolated locations known as Availability Zones. Each Region has three or more Availability Zones.

Availability Zones

Availability Zones are distinct locations within a Region, each designed as an independent zone with its own power, networking, and connectivity. Availability Zones maintain high availability and fault tolerance for applications. Each Availability Zones consists of one or more data centers.

Edge locations

Edge locations are strategically placed sites around the world that cache content to deliver data, video, and applications with lower latency and higher transfer speeds. Edge locations are considered a vital part of the AWS content delivery network (CDN) and use services like CloudFront to efficiently distribute data to end users.

CloudFormation

CloudFormation is a service that helps you model and set up your AWS resources so that you can spend less time managing those resources and more time focusing on your applications that run in AWS. With CloudFormation, you can define your infrastructure as code. You create a template that describes all the AWS resources that you want (like Amazon Elastic Compute Cloud (Amazon EC2) instances), and CloudFormation takes care of provisioning and configuring those resources for you.

Interacting with AWS resources

As you've learned, there are a number of ways you can operate in the AWS Cloud. To interact with AWS resources, you must invoke AWS APIs. To interact with these APIs, you can use the AWS SDKs, the AWS Command Line Interface (AWS CLI), the AWS Management Console, or IaC tools such as CloudFormation

Module 5 - Networking

Networking components

Amazon Virtual Private Cloud (Amazon VPC)

An Amazon VPC lets you provision a logically isolated section of the AWS Cloud where you can launch AWS resources in a virtual network that you define.

Subnet

Subnets are used to organize your resources and can be made publicly or privately accessible. A private subnet is commonly used to contain resources like a database storing customer or transactional information. A public subnet is commonly used for resources like a customer-facing website.

The **AWS Cloud** is the outermost box in most diagrams.

Region is the next box. AWS Regions are separate geographic areas. You choose your Region based on your users' geographic location for lower latency, compliance and data residency requirements, available services, and cost.

Amazon VPC is a solid box, and it represents your isolated, logically segmented network within AWS. A VPC helps you to control your network resources and security.

Availability Zones are shown as separate boxes across a region. AZs consist of one or more discrete data centers, each with redundant power, networking, and connectivity, and housed in separate facilities. Using multiple AZs can protect your applications from the failure of a single location in the Region.

Subnets are essentially segments of your VPC, allowing you to divide your VPC into smaller, manageable sections. A subnet is a range of IP addresses in your VPC.

Private subnets are designed to isolate resources that shouldn't be directly exposed to the public internet. In diagrams, they are illustrated with solid boxes.

Public subnets are designed to provide direct internet access to resources placed inside them. To allow access, they are connected with an internet gateway. You will learn more about internet gateways in a later lesson. In diagrams, public subnets are drawn with dashed boxes.

Amazon VPC

With Amazon VPC, you can provision an isolated section of the AWS Cloud. In this isolated section, you can launch resources in a virtual network that you define. It provides three main benefits. It helps increase security because you can secure and monitor connections, screen traffic, and restrict instance access. Amazon VPC gives you full control over your resource placement, connectivity, and security. The convenience of using Amazon VPC means you will spend less time setting up, managing, and validating your virtual network when compared to on-premises network management.

Subnets

Within an Amazon VPC, you can organize your resources into subsections or subnets. A subnet is a section of an Amazon VPC that can contain resources, such as Amazon EC2 instances. You will learn more about subnets in the next lesson.

Connecting your resources with an internet gateway

To allow public traffic from the internet to access your VPC, you attach an **internet gateway** to the VPC. An internet gateway is a connection between a VPC and the internet. You can think of an internet gateway as being similar to a doorway that customers use to enter the coffee shop. Without an internet gateway, no one can access the resources within your VPC.

Amazon Virtual Private Cloud

Amazon VPC is used to establish boundaries around your AWS resources.

Virtual private gateway

A virtual private gateway allows protected internet traffic to enter into the VPC.

Virtual private network

A VPN encrypts your internet traffic, helping protect it from anyone who might try to intercept or monitor it.

Securely connect a remote workforce to AWS Cloud resources

AWS Client VPN

AWS Client VPN is a networking service you can use to connect your remote workers and on-premises networks to the cloud. It is a fully managed, elastic VPN service that automatically scales up or down based on user demand. Because it is a cloud VPN solution, you don't need to install and manage hardware or try to estimate how many remote users to support at one time.

Benefits: AWS Client VPN provides advanced authentication, remote access. It is elastic and fully managed.

Use case: It can be used to quickly scale remote-worker access.

Securely connect sites to other sites

AWS Site-to-Site VPN

Site-to-Site VPN creates a secure connection between your data center or branch offices and your AWS Cloud resources.

Benefits: Site-to-Site VPN provides high availability, secure and private sessions, and accelerates applications.

Use cases: It can be used for application migration and secure communication between remote locations.

Securely connect resources, even in other VPCs

AWS PrivateLink

AWS PrivateLink is a highly available, scalable technology that you can use to privately connect your VPC to services and resources as if they were in your VPC. You do not need to use an internet gateway, NAT device, public IP address, Direct Connect connection, or AWS Site-to-Site VPN connection to allow communication with AWS services or resources from your private subnets. Instead, you control the specific API endpoints, sites, services, and resources that are reachable from your VPC.

Benefits: AWS PrivateLink helps you secure your traffic and connect with simplified management rules.

Use case: It is used for connecting your clients in your VPC to resources, other VPCs, and endpoints.

Dedicated private connections for increased bandwidth

AWS Direct Connect

Direct Connect is a service that makes it possible for you to establish a dedicated private connection between your network and VPC in the AWS Cloud.

Benefits: AWS Direct Connect reduces network costs and increases amount of bandwidth.

Additional gateway services

AWS Transit Gateway

AWS Transit Gateway is used to connect your Amazon VPCs and on-premises networks through a central hub. As your cloud infrastructure expands globally, inter-Region peering connects transit gateways together using the AWS Global Infrastructure.

Network Address Translation (NAT) Gateway

A NAT gateway is a NAT service. You can use a NAT gateway so that instances in a private subnet can connect to services outside your VPC but external services can't initiate a connection with those instances

Amazon API Gateway

You learned about Application Programming Interface (API)s earlier. Quick refresher, an API defines how different software systems can interact and communicate with each other. The Amazon API Gateway is an AWS service for creating, publishing, maintaining, monitoring, and securing APIs at any scale.

AWS Direct Connect

AWS Direct Connect is a private, dedicated AWS connection to your data center or office.

AWS Client VPN

AWS Client VPN connects your remote workforce to AWS or on-premises with a VPN.

AWS Site-to-Site VPN

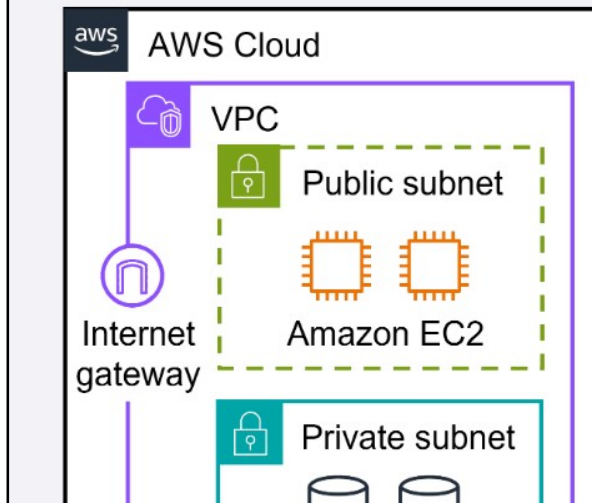
AWS Site-to-Site VPN is an encrypted network connection to your Amazon VPCs.

AWS PrivateLink

AWS PrivateLink connects your VPC privately to services and resources as though they were in your VPC.

Subnets

A section of a VPC for grouping resources based on security or operational needs



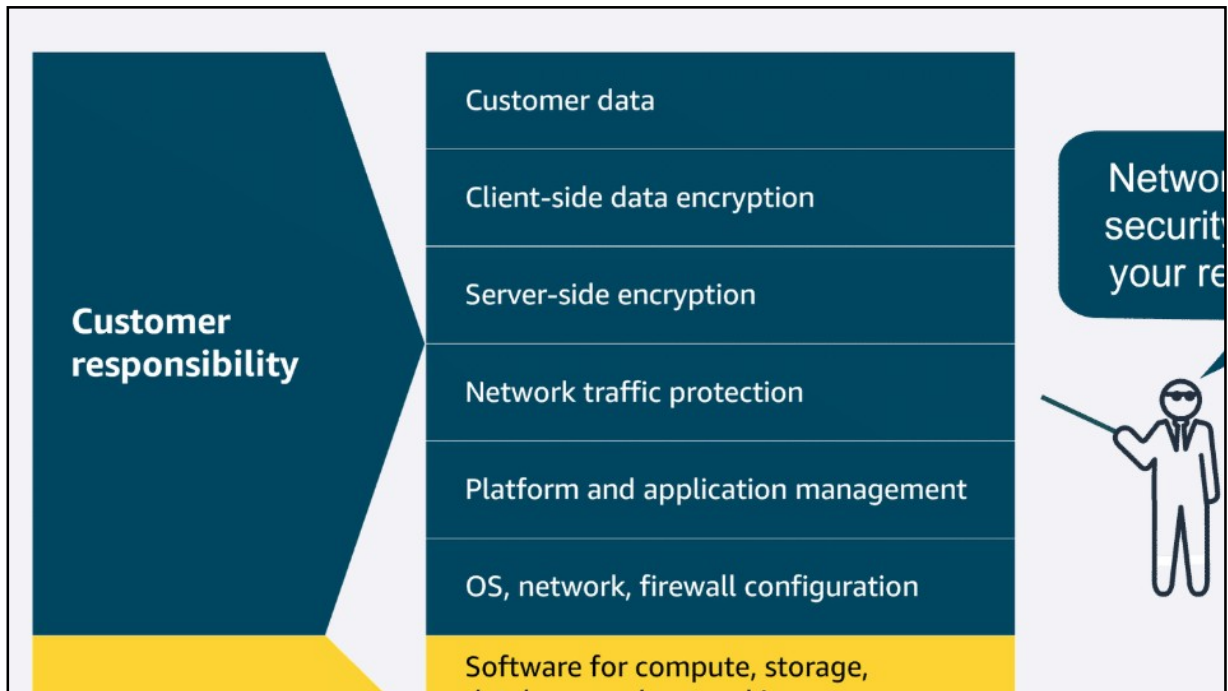
A **subnet** is a section of a VPC in which you group resources based on security or operational needs. Subnets can be public or private.

Public subnets contain resources that need to be accessible from the public, such as an online store's website.

Private subnets contain resources that should not be accessible from the public, such as a database that contains customers' personal information and credit card numbers.

In a VPC, you can define rules to allow resources in one subnet to communicate with resources in another subnet.

Feature	Security Groups	Network ACLs
Scope	Instance level (attached to EC2 instances)	Subnet level (associated with subnets)
State	Stateful (remembers state)	Stateless (doesn't remember state)
Rule types	Only allow type rules	Both allow and deny type rules
Return traffic	Return traffic is automatically allowed if inbound traffic is allowed	Return traffic must be implicitly allowed in both directions
Uses	Fine-grained control of traffic for individual EC2 instances	Broad control of traffic in and out of subnets

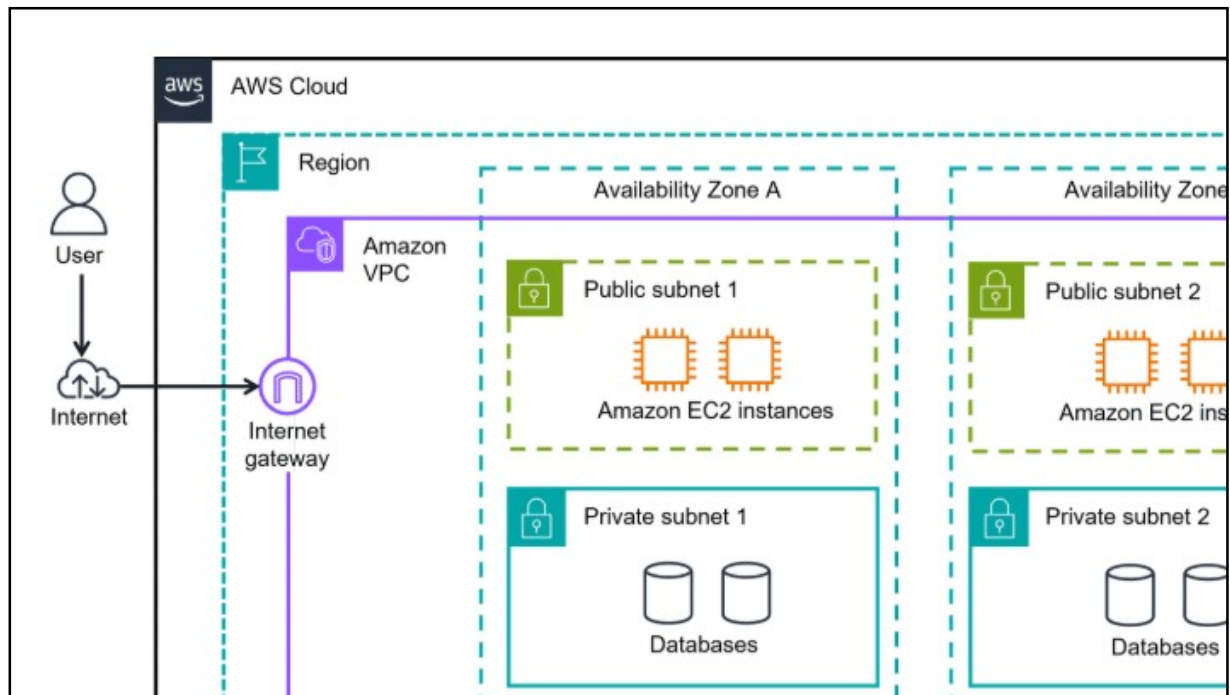


Building an Amazon VPC in the AWS Cloud

Create the Amazon VPC

Create the subnets

Create an internet gateway and route traffic



Edge networking services

Translating domain names to IP addresses with DNS

Suppose that AnyCompany has a website hosted in the AWS Cloud. Customers enter the web address into their browser and they are able to access the website. This happens because of DNS resolution. DNS resolution involves a customer DNS resolver communicating with a company DNS server.

You can think of DNS as being the phone book of the internet. DNS resolution is the process of translating a domain name to an IP address.

Amazon Route 53

Route 53 is a DNS that provides a reliable and cost-effective way to route end users to internet applications.

Route 53 directs end users to your resources with globally dispersed DNS servers and automatic scaling. It gives developers and businesses a reliable way to route end users to internet applications hosted in AWS. It connects user requests to infrastructure running in AWS, such as Amazon EC2 instances and load balancers. It also routes users to infrastructure outside of AWS.

Amazon CloudFront

CloudFront is a content delivery network (CDN) service that delivers your content with faster loading times, cost savings, and reliability. CloudFront is like a global network of delivery trucks that quickly brings web content to users around the world. Instead of all requests traveling back to one central warehouse (your original server), CloudFront stores copies of your content at locations closer to your users. This means websites, videos, images, and applications load much faster, no matter where your customers are located.

AWS Global Accelerator

Global Accelerator is a service that uses the AWS global network to improve application availability, performance, and security. It uses intelligent traffic routing and fast failover if something goes wrong in one of your application locations.

Global Accelerator is a networking service that helps your applications run faster and more reliably across the globe.

Think of it like creating express lanes on the internet highway specifically for your application's traffic. Instead of your users' requests taking the regular, sometimes congested internet routes, Global Accelerator directs traffic through the AWS private global network—getting your users to your application faster and more reliably.

Amazon Route 53

Route 53 is a highly available and scalable cloud DNS service.

Amazon CloudFront

CloudFront is a CDN service that delivers your content with low latency and high speeds.

AWS Global Accelerator

Global Accelerator is a service that uses the AWS global network to improve application availability, performance, and security.

Resource link	Description
Amazon Virtual Private Cloud	Amazon VPC is a service to provision a logically isolated section of the AWS Cloud where you can launch AWS resources in a virtual network that you define.
Subnet	A subnet is a section of a VPC that can contain resources and is used to organize your resources. They can contain be either public or private.
Internet gateway	An internet gateway is a connection between a VPC and the internet. It allows public traffic from the internet to access your VPC.
Virtual private gateway	A virtual private gateway is the component that allows protected internet traffic to enter into the VPC. It allows a connection between your VPC and a private network only if it is coming from an approved network.
AWS Client VPN	Amazon Client VPC is a networking service you can use to connect your remote workers and on-premises networks to the cloud. It is a fully managed, elastic VPN service that automatically scales up or down based on user demand.
AWS Site-to-Site VPN	AWS Site-to-Site VPN creates a secure connection between your data center or branch offices and your AWS Cloud resources.
AWS PrivateLink	AWS PrivateLink is a highly available, scalable technology that you can use to privately connect your VPC to services and resources as though they were in your VPC.
AWS Direct Connect	AWS Direct Connect is a service that provides a dedicated private connection between your data center and a VPC.
Network Access Control List (network ACL)	A network ACL allows or denies specific inbound or outbound traffic at the subnet level using stateless packet filtering.
Security groups	Security groups control the inbound and outbound traffic for a resource at the instance level using stateful packet filtering.
Domain Name System (DNS)	DNS translates human readable domain names to machine readable IP addresses (for example, 192.0.2.0).
Amazon Route 53	Route 53 is a scalable and reliable DNS web service that helps developers and businesses

	route end users to internet applications, whether they're hosted in AWS or elsewhere. It also supports domain registration, health checks, and advanced traffic routing policies.
Amazon CloudFront	CloudFront is a web service that speeds up distribution of your web content to your users through a worldwide network of data centers called edge locations. It securely delivers content with low latency and high transfer speeds.
AWS Global Accelerator	Global Accelerator is a networking service that helps improve the availability and performance of applications for global users by routing traffic through the AWS global network. It helps improve application availability, performance, and security.
Amazon Transit Gateway	Amazon VPC Transit Gateways is a network transit hub used to interconnect VPCs and on-premises networks.
NAT Gateway	Network Address Translation (NAT) gateway allows instances in a private subnet to connect with services outside your VPC. External services can't initiate a connection with those instances.
API Gateway	The Amazon API Gateway is an AWS service for creating, publishing, maintaining, monitoring, and securing APIs at any scale. It handles all the tasks involved in accepting and processing up to hundreds of thousands of concurrent API calls.

Module 6 – Storage

Block storage

Block storage provides persistent, low-latency block-level storage volumes that attach to EC2 instances like physical hard drives. Block storage volumes can be encrypted, backed up via snapshots, and modified while in use without disrupting the instance. AWS offers two primary block storage services:

- *Amazon EC2 instance store*

An unmanaged non-persistent, high-performance block storage directly attached to EC2 instances for temporary data.

- *Amazon Elastic Block Store (EBS)*

A managed service that provides persistent block storage volumes for EC2 instances, offering various types for different workloads

Object storage

Object storage is a data storage architecture that manages data as objects in a flat address space. It offers unlimited scalability so you can store vast amounts of unstructured data without worrying about capacity constraints. Object storage provides enhanced metadata capabilities to provide more efficient data management, search, and analytics across massive datasets.

The following is the primary AWS object storage service:

- *Amazon Simple Storage Service (S3)*

A fully managed scalable object storage service for storing and retrieving any amount of data from anywhere.

File storage

AWS file storage services provide shared file systems accessible over networks, so multiple users and applications can access the same data simultaneously. They offer scalability and flexibility so you can expand storage capacity as needs grow without managing physical infrastructure. AWS offers two primary file storage services:

- *Amazon Elastic File System (EFS)*

A fully managed, scalable NFS file system for use with AWS Cloud services and on-premises resources.

- *Amazon FSx*

A fully managed file storage services for popular file systems like Windows, Lustre, and NetApp ONTAP.

Additional storage services

- *AWS Storage Gateway*

A fully managed, hybrid-cloud storage service that provides on-premises access to virtually unlimited cloud storage.

- *AWS Elastic Disaster Recovery*

A fully managed service that streamlines the recovery of your physical, virtual, and cloud-based servers into AWS.

Resource link	Description
Amazon EC2 Instance Store User Guide	A temporary storage option that is directly attached to the host computer of an EC2 instance, providing high-performance but non-persistent storage.
Amazon Elastic Block Store (Amazon EBS)	A scalable block storage service that provides persistent, high-performance volumes you can attach to your EC2 instances for data storage and applications.
Amazon Elastic Block Store (Amazon EBS) FAQ	Frequently asked questions about Amazon EBS.
Amazon EBS Snapshots User Guide	EBS Snapshots are point-in-time backups of your cloud storage volumes, making it possible to protect data and restore it when needed.
Amazon Data Lifecycle Manager User Guide	A service that streamlines the creation, retention, and deletion of Amazon EBS snapshots.
Amazon Simple Storage Service (Amazon S3)	A scalable cloud storage service that can store and retrieve any amount of data from anywhere on the web.
Amazon Simple Storage Service (Amazon S3) FAQ	Frequently asked questions about Amazon S3.
Amazon S3 Storage Classes	Amazon S3 offers various storage classes, from high-performance frequent access to cost-effective archival options, tailored to different data retrieval needs and budget constraints.
Amazon S3 Versioning User Guide	Amazon S3 versioning keeps multiple variants of objects, offering recovery from unintended deletions or modifications by preserving every update to your files.
Amazon S3 Buckets User Guide	S3 buckets are cloud storage containers that securely hold various types of data, allowing convenient access and management through the AWS online infrastructure.
Amazon Elastic File System (Amazon EFS)	A scalable, fully-managed file storage service that lets multiple AWS resources access shared data simultaneously without capacity planning.
Amazon Elastic File System (Amazon EFS) FAQ	Frequently asked questions about Amazon EFS.
Amazon FSx	A fully managed file storage service that lets you launch and run file systems like Windows File Server, Lustre, NetApp ONTAP, and OpenZFS in the AWS cloud.
Amazon FSx for Windows File Server	An Amazon FSx option providing reliable, high-performance file storage compatible with Windows applications in the AWS Cloud.
Amazon FSx for NetApp ONTAP	An Amazon FSx option providing file storage with advanced data management capabilities and compatibility with both Windows and Linux workloads on AWS.
Amazon FSx for OpenZFS	An Amazon FSx option that provides high-performance, scalable storage using the popular open-source ZFS file system.
Amazon FSx for Lustre	An Amazon FSx option designed to accelerate workloads by providing fast data access for compute-intensive applications in AWS.
AWS Storage Gateway	A hybrid cloud storage service that provides seamless and secure integration between on-premises environment and AWS

	cloud storage services.
Amazon S3 File Gateway	A Storage Gateway configuration that provides local file access to S3 objects while caching frequently accessed data locally for faster retrieval.
Tape Gateway	A Storage Gateway configuration used for backing up data to Amazon S3 while maintaining compatibility with existing tape-based backup applications.
Volume Gateway	A Storage Gateway configuration that provides iSCSI block storage volumes to on-premises applications, offering both cached and stored modes.

Module 7 – Databases

Relational databases

Amazon Relational Database Service (Amazon RDS)

Supported database engines

Amazon RDS supports different database engines, including Amazon Aurora, MySQL, PostgreSQL, Microsoft SQL Server, MariaDB, and Oracle Database.

Use cases

Some examples of practical use cases for Amazon RDS are web applications, enterprise workloads, and product inventories for e-commerce platforms.

Benefits

Cost optimization

Multi-AZ deployment

Performance optimization

Security controls

Amazon Aurora

Aurora is a managed relational database designed to help reduce unnecessary I/O operations. It's compatible with MySQL and PostgreSQL, provides high performance and availability, and automatically scales alongside your workloads. Aurora replicates data across multiple Availability Zones for enhanced durability and fault tolerance, and features automated backups, encryption at rest, and continuous monitoring.

Use cases

Some examples of practical use cases for Aurora are gaming applications, media and content management, and real-time analytics.

Benefits

High performance and availability

Automated storage and backup management

Advanced replication and fault tolerance

NoSQL databases

Amazon DynamoDB

DynamoDB is a fully managed NoSQL database service that provides fast and predictable performance for both document and key-value data structures. It's a powerful and incredibly fast database option for use cases that require a flexible schema, and is ideal for applications that require high performance and seamless scaling.

DynamoDB seamlessly scales alongside your data without impacting performance, which means that you only pay for the resources that you use. It also includes built-in security features for enhanced protection, and automatically spreads your data across multiple servers to handle your workload.

Use cases

Some examples of practical use cases for DynamoDB are gaming platforms, financial service applications, and mobile applications with global user bases.

Benefits

Scalability with provisioned capacity

Consistent high performance

High availability and durability

Data encryption

In-memory caches

Amazon ElastiCache

ElastiCache is a fully managed in-memory caching service that was built to help reduce the complexity of administering in-memory caching systems. This means that you can continue to use the same Redis, Valkey, or Memcached tools and configurations to scale your workloads. It automatically detects and replaces failed nodes, which makes it ideal for applications that need consistent high performance.

Use cases

Some examples of practical use cases for ElastiCache are session data management, database query enhancement, and gaming leaderboards.

Benefits

High performance for Redis, Valkey, or Memcached instances

High availability

Replication across multiple Availability Zones

Data encryption

Additional DB Services

Amazon DocumentDB

AWS Backup

Amazon Neptune

Amazon Relational Database Service (Amazon RDS)	A relational database service supporting multiple engines like MySQL, PostgreSQL, and Microsoft SQL Server with automated maintenance and backups
Amazon RDS Security	Detailed information about security configurations in Amazon RDS
Amazon Aurora	A cloud-native database offering superior performance and availability over traditional databases while maintaining MySQL and PostgreSQL compatibility
AWS Database Migration Service (AWS DMS)	A service that provides seamless database migration between source and target databases while keeping the source database operational
Amazon DynamoDB	A NoSQL database service providing single-digit millisecond performance at any scale with built-in security
Amazon ElastiCache	An in-memory caching service that supports Redis, Valkey, or Memcached to improve application performance through faster data retrieval
Amazon DocumentDB	A MongoDB-compatible document database service designed for mission-critical workloads with automatic scaling
Amazon Backup	A centralized service for automating and managing data backups across AWS services and on-premises resources
Amazon Neptune	A graph database service optimized for storing and querying highly connected data relationships
What Is a Relational Database?	A structured database using tables with predefined schemas, supporting complex queries and transactions through SQL for consistent data relationships
What Is a NoSQL Database?	A nonrelational database offering flexible schemas and high

	scalability for varied data types, optimized for specific data models and patterns
What Is an In-Memory Caching Service?	A high-speed data storage layer using RAM instead of disk storage, delivering microsecond latency for frequently accessed data
AWS Shared Responsibility Model	AWS is responsible for security of the cloud (infrastructure, hardware, networking, facilities) while customers are responsible for security in the cloud (data, configuration, access management).

Module 8 – AI, ML & Analytics

Amazon Comprehend	Use natural language processing to extract key insights from documents.
Amazon Polly	Convert text into lifelike speech.
Amazon Transcribe	Convert speech into text.
Amazon Translate	Translate text into multiple languages.
Amazon Kendra	Use natural language processing to intelligently query enterprise content.
Amazon Rekognition	Identify objects and activities in images and videos.
Amazon Textract	Detect and extract typed and handwritten text in documents.
Amazon Lex	Add voice and text conversational interfaces to applications.
Amazon Personalize	Add personalized customer recommendations to applications.
Amazon SageMaker AI	Build, train, and deploy your own ML models without worrying about infrastructure.
Amazon SageMaker JumpStart	Deploy pre-trained ML solutions, like computer vision, NLP, and tabular data, with just a few clicks.
Amazon Bedrock	Fine-tune and seamlessly integrate large FMs from Amazon and leading AI startups into your AWS applications with a single API.
Amazon Q Business	Answer questions and solve problems using the data and expertise found in your company's information repositories.
Amazon Q Developer	Accelerate development with code recommendations.
Amazon Kinesis Data Streams	Ingest terabytes of streaming data in real time.
Amazon Data Firehose	Ingest and deliver data within seconds to multiple consumers, such as data lakes and warehouses.
Amazon S3	Store virtually limitless amounts of all types of data using this popular data lake choice.
Amazon Redshift	Store petabytes of structured or semistructured data in this data warehouse service. Analyze large datasets stored in the warehouse using complex SQL queries.
AWS Glue Data Catalog	Provide metadata to various analytics services with this centralized repository.
AWS Glue	Process data by using the AWS Glue Data Catalog as a reference.
Amazon EMR	Process big data workloads by using popular frameworks like Apache Spark.
Amazon Athena	Analyze various data sources hosted anywhere with a single SQL query.
Amazon QuickSight	Visualize data by creating interactive dashboards with or without expertise.
Amazon OpenSearch Service	Visualize and monitor real-time data analytics with keyword or NLP searches.

Module 9 – Security

AWS shared responsibility model

Cloud security is a shared responsibility between customers and AWS. Let's examine this relationship using the AWS shared responsibility model.

Customers: Security *in* the cloud

When using AWS services, customers maintain complete control over their content. As a result, customers are responsible for securing everything they create and manage in the AWS Cloud. This includes the following:

- Managing the security of data, systems, and applications
- Deciding what data and workloads to store or run in AWS
- Determining which AWS services to use
- Controlling who has access to environments and resources

AWS: Security *of* the cloud

AWS is responsible for security *of* the cloud. AWS operates, manages, and controls the components at all layers of the infrastructure. This includes securing the following:

- The foundational software that powers AWS services
- The virtualization layer
- The hardware and global infrastructure that supports the data centers from which services operate. This includes protection for AWS Regions, Availability Zones, and edge locations.

AWS Identity and Access Management (IAM)	Securely manage identities and access to AWS services and resources.
AWS IAM Identity Center	Connect your existing workforce identity source and centrally manage access to AWS with single sign-on.
AWS Secrets Manager	Centrally store and manage credentials, API keys, and other secrets.
AWS Systems Manager	Manage nodes, or connection points, at scale on AWS and in multi-cloud and hybrid environments.
AWS Shield	Protect your network and applications from the most common, frequently occurring types of DDoS attacks.
AWS WAF	Protect your network and applications from blocked IP addresses defined by a web ACL.
AWS Key Management Service (AWS KMS)	Create and manage cryptographic keys to encrypt and decrypt your data.
Amazon Macie	Certify that sensitive data is discovered and protected in Amazon S3.
AWS Certificate Manager (ACM)	Create and manage SSL/TLS certificates that provide data encryption in transit.
Amazon Inspector	Check applications for security vulnerabilities and deviations from security best practices.
Amazon GuardDuty	Continuously monitor the AWS environment with intelligent threat detection.
Amazon Detective	Analyze threats with interactive visualizations contained in a unified view.

AWS Security Hub	Aggregate security findings and organize them into actionable insights.
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Module 10 – Monitoring, Compliance & Governance

Resource link	Description
Amazon CloudWatch	CloudWatch monitors your AWS resources and the applications you run on AWS in real time. With CloudWatch, you gain system-wide visibility into resource utilization, application performance, and operational health.
AWS CloudTrail	CloudTrail enables auditing, security monitoring, and operational troubleshooting. CloudTrail records user activity and API calls across AWS services as events. CloudTrail events help you answer the question of "Who did what, where, and when?"
AWS Artifact	AWS Artifact a self-service portal that provides on-demand access to AWS security and compliance documentation, including reports, certifications, and agreements.
AWS Config	AWS Config is a service to assess, audit, and evaluate the configurations of your AWS resources.
AWS Audit Manager	Audit Manager is a service to continually audit your AWS usage to streamline risk and compliance assessment.
AWS Organizations	Organizations helps you centrally manage and govern your environment as you grow and scale your AWS resources. It helps you manage policies for groups of accounts and automate account creation.
AWS Control Tower	With AWS Control Tower, you can enforce and manage governance rules for security, operations, and compliance at scale across all your organizations and accounts in the AWS Cloud.
AWS Service Catalog	With Service Catalog, you can create, share, and organize from a curated catalog of AWS resources. You can deploy baseline networking resources and security tools for new AWS accounts so you can govern consistently.
AWS License Manager	License Manager is a service that helps you manage your software licenses and fine-tune your licensing costs.
AWS Trusted Advisor	Trusted Advisor helps you optimize costs, increase performance, improve security and resilience, and operate at scale in the cloud. It continuously evaluates your AWS environment using best practice checks across those categories and recommends actions to remediate deviations from best practices.
AWS Health	AWS Health is the data source for events and changes affecting your AWS Cloud resources. AWS Health notifies you about service events, planned changes, and account notifications to help you manage and take actions.
AWS Identity and Access Management Access Analyzer	IAM Access Analyzer provides capabilities to set, verify, and refine security permissions to achieve least privilege security standards.

Module 11 – Pricing and Support

Key concepts of AWS pricing

Pay as you go

With pay as you go, you can adapt to changing business needs and reduce the risk of overprovisioning or missing capacity.

Save when you commit

For certain services, such as Compute services on AWS, Savings Plans offer savings over On-Demand prices when you commit to a 1-year or 3-year plan.

Pay less by using more

With AWS, you can realize important savings as your usage increases. For some services, pricing is tiered, meaning the more you use, the less you pay.

Driving factors of cost

Compute

For compute resources, you pay by a certain span of time, like by the hour or by the second. Unless you've made a reservation for which the cost is agreed upon beforehand, you pay from the time you launch a resource until the time you stop the instance.

Storage

You can choose from a broad portfolio of storage solutions with deep functionality for storing, accessing, protecting, and analyzing data. Pricing for storage largely depends on how much storage you have provisioned or how much you are using.

Data transfer

In most cases, there is no charge for inbound data transfer or for data transfer between AWS services within the same Region. There are some exceptions, so be sure to verify data transfer rates before beginning.

Outbound data transfer is aggregated across services and then charged at the outbound data transfer rate. The more data you transfer, the less you pay per gigabyte. For data storage and transfer, you typically pay per gigabyte.

AWS pricing and billing services

AWS Organizations

AWS Organizations provides centralized management and governance of your AWS environment. Using AWS Organizations, you can create, group, and manage accounts. You can

also apply security policies at the account level and consolidate billing with multiple accounts using a single payment method.

Use cases:

- Consolidate multiple AWS accounts into one central organization.
- Implement organization-wide policies.

AWS Billing and Cost Management dashboard

The AWS Billing and Cost Management dashboard centralizes cost management, showing current charges, usage, forecasts, and detailed breakdowns. It also provides tools to manage payments, view invoices, set budgets, and consolidate billing.

Use cases:

- Use helpful visualizations and billing reports of monthly AWS spend.
- Set up and manage payment methods.

AWS Budgets

AWS Budgets helps set custom budgets and sends alerts when costs, usage, or Savings Plans and Reserved Instances (RIs) utilization or coverage exceed defined thresholds.

Use cases:

- Set up alerts for when projected costs exceed predefined thresholds.
- Forecast future expenses based on current usage trends.

AWS Cost Explorer

AWS Cost Explorer helps visualize, analyze, and manage AWS costs and usage with interactive graphs, reports, and forecasts. It provides insights into spending patterns, trends, and Reserved Instance recommendations.

Use cases:

- Analyze historical spending trends to identify cost-saving opportunities.
- Forecast future AWS costs based on current usage patterns to budget effectively.

AWS Pricing Calculator

Another helpful tool is the AWS Pricing Calculator. The AWS Pricing Calculator is a web-based planning tool that you can use to create estimates. You input specific configurations such as instance types, storage options, and data transfer volumes. Then, based on your configurations, you receive a detailed cost breakdown to help you budget for your AWS resource allocation.

Use cases:

- Estimate potential costs before deployment.
- Compare costs of different AWS services and configurations.

Types of AWS support

Basic Support	Developer Support *	Business Support *	Enterprise On-Ramp Support *	Enterprise Support
Included for all AWS customers	Recommended for experimenting or testing in AWS	Recommended minimum tier for production workloads in AWS	Recommended for production and business critical workloads in AWS	Recommended for business critical and mission critical workloads in AWS
Includes access to documentation, whitepapers, and AWS re:Post	Response times: • < 24 hours for general guidance • < 12 hours when systems impaired	Response times: • <i>Includes previous plan response times</i> • < 4 hours when production system impaired • < 1 hour when production system is down	Response times: • <i>Includes previous plan response times</i> • < 30 minutes when business-critical system is down	Response times: • <i>Includes previous plan response times</i> • < 15 minutes when business- or mission-critical system is down
Core AWS Trusted Advisor checks	Core AWS Trusted Advisor checks	Full set of AWS Trusted Advisor checks	Full set of AWS Trusted Advisor checks	Full set of AWS Trusted Advisor checks and prioritized recommendations by AWS account team
Technical Account Management not included	Technical Account Management not included	Technical Account Management not included	A pool of technical account managers (TAMs) provide proactive guidance	A designated TAM provides consultative architectural and operational guidance

AWS Marketplace

Software as a service (SaaS):

- Business applications such as project management tools
- Marketing tools such as customer engagement platforms
- Collaboration tools such as file sharing services

Machine learning (ML) and AI:

- Prebuilt models for image recognition, natural language processing, and more
- ML algorithms for training custom models

Data and analytics:

- Business intelligence platforms for visualization and reporting
- Data integration tools

AWS Partner Network

The AWS Partner Network (APN) is a global community that uses AWS technologies, programs, expertise, and tools to build solutions and services for customers. Together, partners and AWS provide innovative solutions, solve technical challenges, and deliver customer value. You can work with AWS Partners to create or use specialized solutions that are tailored to your unique business needs.

Funding benefits: As businesses join the AWS Partner Network and participate in specific programs available to AWS Partners, they can unlock various funding benefits to help build, market, and sell with AWS.

AWS Partner events: AWS Partner events include webinars, virtual workshops, and in-person learning opportunities. You can use AWS Partner events to network with other partners, learn more about new or current offerings, and collaborate with AWS experts.

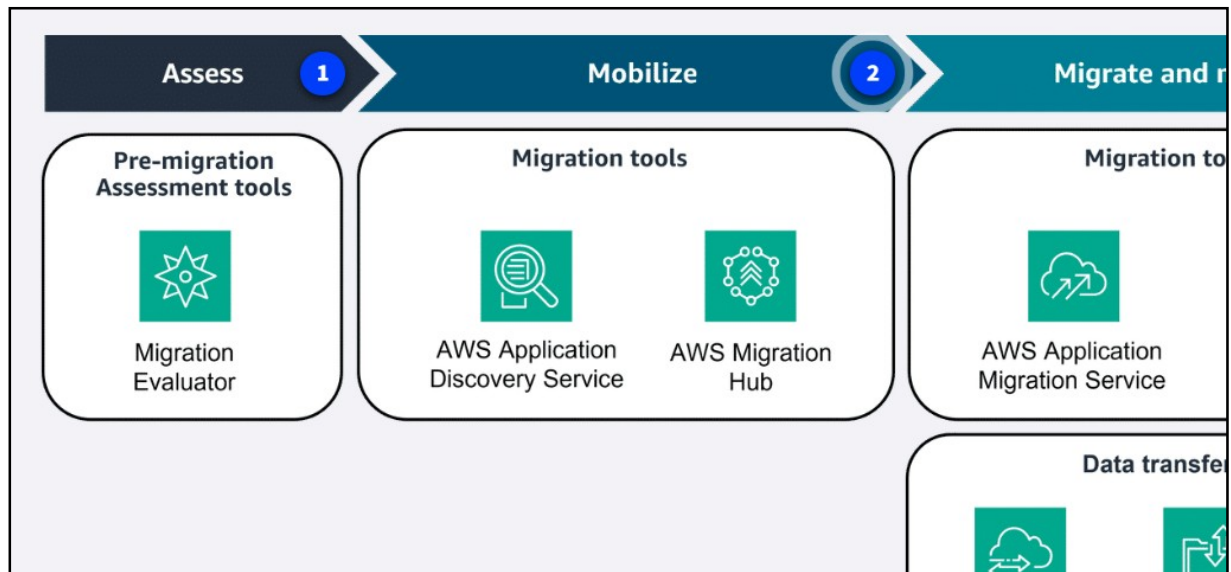
AWS Partner Training and Certification: Take advantage of unique, partner-centered offerings from AWS Training and Certification. From certification to a specific service or learning objective, the AWS Partner training portfolio has numerous opportunities to upskill your cloud knowledge.

Resource link	Description
Amazon S3 Pricing	The Amazon S3 Pricing page outlines costs for storage, requests, data transfer, and additional features, along with details on AWS Free Tier for new customers.
Amazon EC2 On-Demand Pricing	The Amazon EC2 On-Demand Pricing page details the costs for EC2 instances, including pricing for instance types, data transfer, and additional features with no long-term commitments.
What is AWS Organizations?	The AWS Organizations User Guide explains how to centrally manage multiple AWS accounts, automate account creation, apply governance policies, and simplify billing and resource sharing.
AWS Billing Console	The AWS Billing Console provides a centralized platform for managing and understanding AWS charges, offering tools to view and download invoices, monitor discounts and credits, and analyze spending trends across multiple accounts.
AWS Budgets	AWS Budgets provides tools to create custom budgets for tracking and managing AWS costs, usage, and Reserved Instance or Savings Plan use. It also helps prevent overspending by offering alerts and automated actions.
AWS Cost Explorer	AWS Cost Explorer provides tools to visualize, analyze, and manage AWS costs and usage, with customizable reports and cost forecasting to optimize cloud spending.
AWS Pricing Calculator Documentation	The AWS Pricing Calculator is a free, web-based tool that helps customers create detailed cost estimates for AWS services, so they can plan and manage their cloud expenditures effectively.
AWS re:Post	AWS re:Post is a community-driven, question-and-answer platform provided where users can seek help, share knowledge, and find solutions related to AWS services and technologies.
AWS Trust and Safety Center	The AWS Trust and Safety Center offers guidance on reporting abuse, protecting applications, and following best practices for digital messaging on AWS.
AWS Professional Services	The AWS Professional Services page highlights how AWS experts help organizations

	accelerate cloud adoption with tailored guidance across industries and technologies.
Welcome to AWS Documentation	The AWS Documentation page provides comprehensive technical resources, including guides, API references, tutorials, and best practices across AWS services.
AWS Marketplace	The AWS Marketplace is a digital catalog where customers can discover, purchase, and manage third-party software, data, and services with flexible pricing options.
Engage with AWS Partners	The AWS Partners page helps you discover how AWS Partners can help you accelerate your cloud journey.
Funding Benefits for AWS Partners	The AWS Partner Funding page outlines funding benefits, including credits and discounts, to support AWS partners in developing, marketing, and selling solutions based in AWS.
AWS Partner Events	The AWS Partner Events page lists upcoming webinars, workshops, and in-person events for IT professionals, developers, and business leaders, with options to filter by date, location, and topic.
AWS Partner Training and Certification	The AWS Partner Training page offers courses, certifications, and resources to help AWS Partners build cloud skills across various roles and enhance their expertise.

Module 12 – Migrating to AWS Cloud

Three phases of the migration process



Improve your cloud readiness

AWS CAF

The AWS CAF is a framework that brings AWS experience and best practices to companies preparing to migrate to the AWS Cloud. The framework provides tools to help accelerate the migration journey, organize resources, and align management during the transition.

Functional and business-related stakeholders



Business

The Business perspective makes sure that IT aligns with business needs and that IT investments link to key business results.

Use the Business perspective to create a strong business case for cloud adoption and prioritize cloud adoption initiatives. Make sure that your business strategies and goals align with your IT strategies and goals.

The following are common Business perspective roles:

- Business managers
- Finance managers
- Budget owners
- Strategy stakeholders

People

The People perspective supports development of an organization-wide change management strategy for successful cloud adoption.

Use the People perspective to evaluate organizational structures and roles, assess new skill and process requirements, and identify gaps. This helps prioritize training, staffing, and organizational changes.

The following are common People perspective roles:

- Human resources
- Staffing
- People managers

Governance

The Governance perspective focuses on skills and processes to align IT strategy with business strategy. This perspective helps you maximize business value and minimize risks.

Use it to understand how to update the staff skills and processes necessary to maintain business governance in the cloud. Manage and measure cloud investments to evaluate business outcomes.

The following are common Governance perspective roles:

- Chief information officer (CIO)
- Program managers
- Enterprise architects
- Business analysts
- Portfolio managers

Platform

The Platform perspective includes principles and patterns for implementing new solutions in the cloud and migrating on-premises workloads to the cloud.

Use a variety of architectural models to understand and communicate the structure of IT systems and their relationships. Describe the architecture of the target state environment in detail.

The following are common Platform perspective roles:

- Chief technology officer (CTO)
- IT managers
- Solutions architects

Security

The Security perspective makes sure that the organization meets security objectives for visibility, auditability, control, and agility.

Use AWS CAF to structure the selection and implementation of security controls that meet the organization's needs.

The following are common Security perspective roles:

- Chief information security officer (CISO)
- IT security managers
- IT security analysts

Operations

The Operations perspective helps you to enable, run, use, operate, and recover IT workloads to the level agreed upon with your business stakeholders.

Define how day-to-day, quarter-to-quarter, and year-to-year business is conducted. Align with and support the operations of the business. AWS CAF helps these stakeholders define current operating procedures and identify the process changes and training needed to implement successful cloud adoption.

The following are common Operations perspective roles:

- IT operations managers
- IT support managers

Seven migration strategies

Relocate

Relocating is changing the hosting location to the cloud. This could be if applications are already virtual machines (VMs) or containers running on premises and then moving to the cloud.

Rehost

Rehosting, also known as lift-and-shift, involves moving applications without changes.

In a scenario of a large legacy migration, in which the company is looking to implement its migration and scale quickly to meet a business case, the majority of applications are rehosted.

Replatform

Replatforming, also known as lift, tinker, and shift, involves making a few cloud optimizations to realize a tangible benefit. Optimization is achieved without changing the core architecture of the application.

Refactor

Refactoring, also known as re-architecting, involves reimagining how an application is architected and developed by using features built for the cloud. Refactoring is driven by a strong business need to add features, scale, or improve performance that would otherwise be difficult to achieve in the application's existing environment.

Repurchase

Repurchasing involves moving from a traditional license to a software-as-a-service (SaaS) model.

For example, a business might choose to implement the repurchasing strategy by migrating from a customer relationship management (CRM) system to a new sales force software.

Retain

Retaining consists of keeping applications that are critical for the business in the source environment. This might include applications that require major refactoring before they can be migrated or work that can be postponed until a later time.

Retire

Retiring is the process of removing applications that are no longer needed.

Migrating databases**AWS DMS**

The AWS Database Migration Service (AWS DMS) makes it possible to quickly and securely migrate databases and perform ongoing data replication tasks for live databases and data

warehouses. It provides a way to plan, assess, convert, and migrate databases even with data warehouses in one central tool.

AWS SCT

AWS SCT makes it possible to convert database schemas and code objects (like stored procedures, views, and functions) from one database engine to another. AWS SCT can even give you estimates of how big of an effort a conversion is, which helps with planning.

Transferring data to and from the AWS Cloud (Online)

AWS DataSync

AWS DataSync is specifically designed for automating and accelerating data transfer. DataSync simplifies and accelerates moving large amounts of data between on-premises storage and AWS storage services like Amazon Simple Storage Service (Amazon S3). It automates many aspects of the transfer process, including running instances, encryption, and network optimization.

AWS Transfer Family

The AWS Transfer Family makes it possible to seamlessly manage and share data with simple, secure, and scalable file transfers. This service provides fully managed support for secure file transfers over FTP, Secure File Transfer Protocol (SFTP), File Transfer Protocol Secure (FTPS), and other protocols. It helps you transfer files directly into and out of AWS storage services like Amazon S3 and Amazon EFS.

Direct Connect

AWS Direct Connect is a service that makes it possible for you to establish a dedicated private connection between your network and virtual private cloud (VPC) in the AWS Cloud. Because it is your dedicated connection, it is a fast, reliable, and secure way to transfer your data or files.

Alternatives for data migration (Offline)

Snowball Edge Storage Optimized devices

AWS Snowball Edge Storage Optimized devices are a great solution for offline data migration where connecting to the internet might not be an option. These devices deliver high performance NVMe storage, making it possible to simplify multi-petabyte data migrations from on-premises locations to AWS.

Resource link	Description
Three Phases of Migration	In performing a large-scale migration, there are typically three phases: assess, mobilize, and the last phase, migrate and modernize.
AWS Cloud Adoption Framework	AWS CAF is a framework that brings AWS experience and best practices to companies preparing to migrate to the AWS Cloud. The framework provides tools to help accelerate the migration journey, organize resources, and align management during the transition.

Seven Migration Strategies (7 Rs)	When migrating to the cloud, there are seven common migration strategies that customers can choose to implement. They are relocate, rehost, replatform, refactor, repurchase, retain, and retire.
Migration Evaluator	The Migration Evaluator is a migration assessment service that helps you create a business case for AWS Cloud planning and migration. It does this with a data-driven approach, analyzing your current state and target state, and developing a migration readiness plan with projected cloud costs.
AWS Application Discovery Service	Application Discovery Service discovers on-premises server inventory and connections. It gathers configuration, performance, and connection details for both servers and databases to create a detailed migration plan.
AWS Application Migration Service	Application Migration Service is a tool to migrate and improve your on-premises and cloud-based applications. It helps customers simplify, expedite, and reduce the cost of migrating and modernizing applications.
AWS Migration Hub	Migration Hub is a centralized hub to take you from discovery to assessment, planning, and implementation of your migration. It provides tools, guidance, and automated recommendations to collaborate with your team and track your migration.
AWS Database Migration Service (AWS DMS)	AWS DMS makes it possible to quickly and securely migrate databases to the AWS Cloud. It provides a way to plan, assess, convert, and migrate databases even with data warehouses in one central tool.
AWS Schema Conversion Tool (AWS SCT)	AWS SCT is a service that makes it possible to convert database schemas and code objects (like stored procedures, views, and functions) from one database engine to another.
AWS DataSync	DataSync is specifically designed for automating and accelerating secure data migrations. DataSync manages data movement workloads with bandwidth throttling, migration scheduling, task filtering, and task reporting. It also provides rapid data replication.
AWS Transfer Family	Transfer family makes it possible to seamlessly manage and share data with simple, secure, and scalable file transfers. This service provides fully managed support for secure file transfers over SFTP, FTPS, FTP, and others. It helps you transfer files directly into and out of AWS storage services like Amazon S3 and Amazon EFS.
AWS Direct Connect	Direct Connect is a cloud service that you can use to establish a private, dedicated network connection between your on-premises network and AWS.
AWS Snow Family	AWS Snowball Edge Storage Optimized devices are a great solution for offline data migration where connecting to the internet might not be an option. Snowball Edge Storage Optimized devices deliver high performance storage, and make it possible to simplify multi-petabyte data migrations from on-premises locations to AWS.

Module 13 – Well-Architected Solutions

Types of services

Development services

AWS offers several services to help developers automate CI/CD pipelines, monitor and debug applications, build

GraphQL APIs, and deploy web and mobile applications on AWS. Let's examine these services in a bit more detail.

AWS CodeBuild

CodeBuild is a fully managed continuous integration service that compiles source code, runs tests, and produces software packages for deployment. It automatically scales to meet demand, and you only pay for the build time that you use.

AWS CodePipeline

CodePipeline is a fully managed CI/CD service that automates the build, test, and deploy phases of your release process. This helps developers streamline software release workflows and reliably deliver new features and fixes without needing to provision servers.

AWS X-Ray

X-Ray is a powerful tracing, debugging, and performance analysis tool that helps developers visualize application behavior. With X-Ray, developers can quickly identify performance bottlenecks, troubleshoot issues, and optimize applications for better efficiency and reliability.

AWS AppSync

AWS AppSync is a fully managed GraphQL service. With AWS AppSync, developers can create a single GraphQL API that can securely access, manipulate, and combine data from multiple data sources. This helps developers connect frontend applications with backend data.

AWS Amplify

Amplify helps you streamline the process of developing, deploying, and managing secure and scalable full-stack applications on AWS. With Amplify, developers can quickly add features, like authentication, APIs, storage, and hosting, with minimal infrastructure management.

Business application services**Amazon Connect**

Businesses can use this AI-powered contact center service to efficiently set up and operate a scalable customer service call center. Amazon Connect provides capabilities for call routing, recording, and analytics while integrating seamlessly with other AWS services.

Amazon Simple Email Service (Amazon SES)

Amazon SES is a scalable and cost-effective email service provider that can be integrated into any application for reliable, high-volume email automation. It helps businesses optimize the delivery of transactional and marketing emails, resulting in enhanced customer engagement.

End-user computing services**Amazon AppStream 2.0**

AppStream 2.0 is a fully managed service that streams applications from the cloud directly to any compatible device. This includes software-as-a-service (SaaS) applications and applications converted from desktop to SaaS without code revisions. This provides instant access to powerful software without the need for high-end local hardware.

Amazon WorkSpaces

WorkSpaces is a fully managed cloud-based desktop computing service. With WorkSpaces, employees can securely access their work environment from any device with an internet connection. Employees can perform the same tasks as if they were on a physical office computer, while companies can benefit from cost-efficiency and easy administration.

Amazon WorkSpaces Secure Browser (formerly Amazon WorkSpaces Web)

WorkSpaces Secure Browser is a fully managed remote enterprise browser. It provides a protected environment for employees to access private websites, SaaS applications, and the public internet. With WorkSpaces Secure Browser, IT departments don't need to manage specialized client software, infrastructure, or VPN connections.

IoT services

AWS IoT Core

AWS IoT Core is a managed cloud service used to securely connect physical devices with cloud applications. It helps you create efficient IoT solutions by streamlining the complex process of ingesting, processing, and acting on device data. Device connections and data are secured with mutual authentication and end-to-end encryption, and you can choose from several communication protocols.

Well-Architected Framework pillars

Operational Excellence:

Focuses on operations, monitoring, automation, and continuous improvement

Security:

Protects systems and data through best practices like least privilege and data integrity

Reliability:

Emphasizes recovery planning and system adaptability to meet changing demands

Performance Efficiency:

Encourages using the right resources for the job and adjusting as needs evolve

Cost Optimization:

Helps control and reduce costs through smart provisioning and resource management

Sustainability:

Promotes energy-efficient design and environmentally conscious resource usage

Resource link	Description
AWS CodeBuild	Build scripts for compiling, testing, and packaging your code.
AWS CodePipeline	Automate and monitor CI/CD pipelines.
AWS X-Ray	Monitor and debug applications.
AWS AppSync	Connect applications to multiple data sources using a single GraphQL API request.
AWS Amplify	Quickly develop, deploy, and manage full-stack applications on AWS.
Amazon Connect	Manage customer service operations with artificial intelligence.
Amazon SES	Use automation to send large volumes of transactional and marketing emails.
Amazon AppStream 2.0	Stream cloud-based and desktop applications to users on compatible devices.
Amazon WorkSpaces	Provide employees with secure virtual desktops.
Amazon WorkSpaces Secure Browser	Provide employees with secure access to web-based applications and sites.
AWS IoT Core	Securely connect devices with cloud applications to create IoT solutions.
AWS Well-Architected	The AWS Well-Architected Framework provides a structured approach to designing and operating secure, high performing, resilient, and efficient infrastructure for applications and workloads on AWS, guided by six key pillars.
AWS Well-Architected Framework	This website is the official documentation for the AWS Well-Architected Framework.
What is AWS Well-Architected Tool?	The AWS Well-Architected Tool provides a consistent process for measuring your architecture using AWS best practices.
AWS Architecture Center	The AWS Architecture Center provides reference architectures and best practices for building secure, efficient, and scalable solutions on AWS.
AWS Architecture Blog	The AWS Architecture Blog shares technical insights and best practices for building effective solutions on AWS.